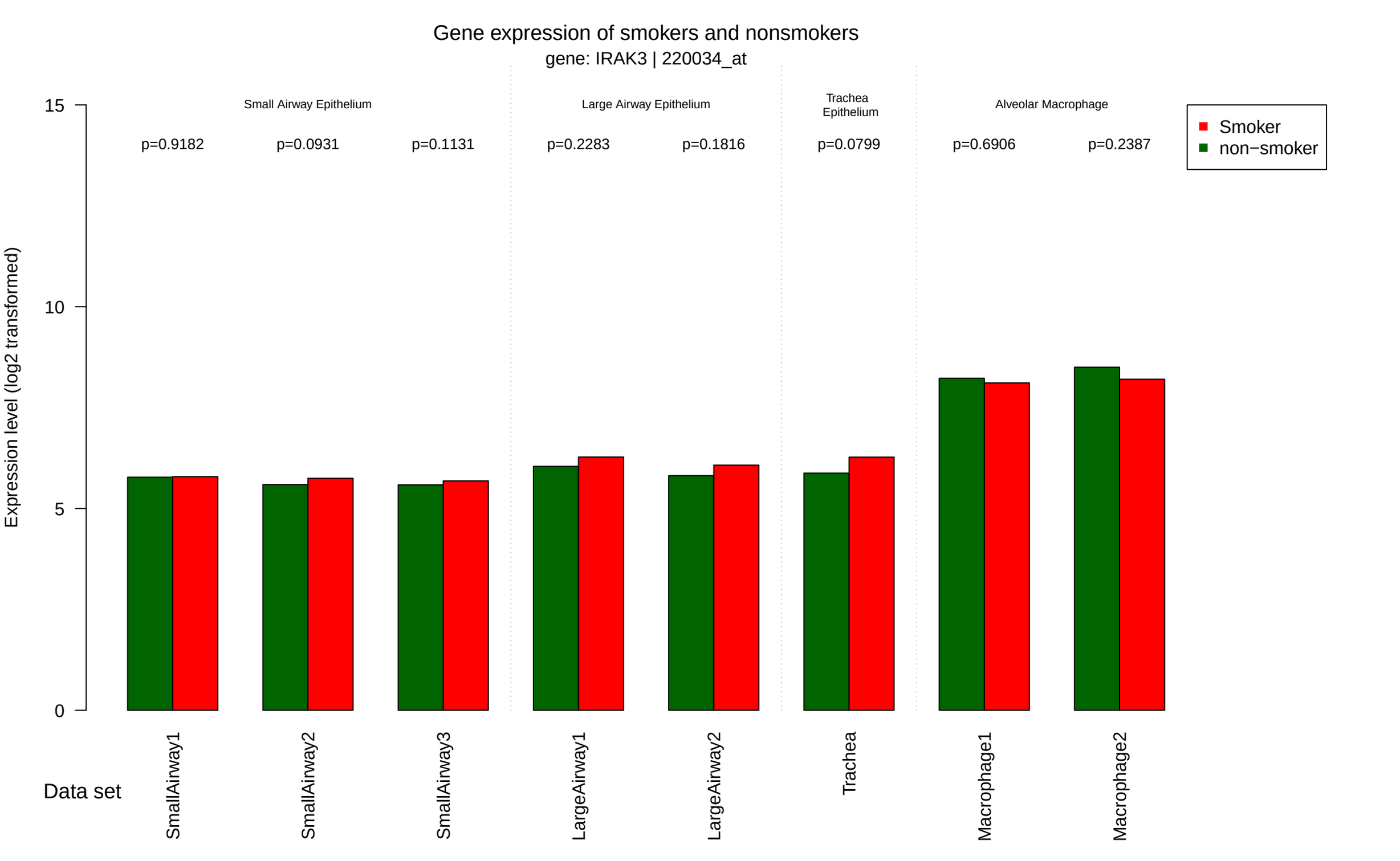
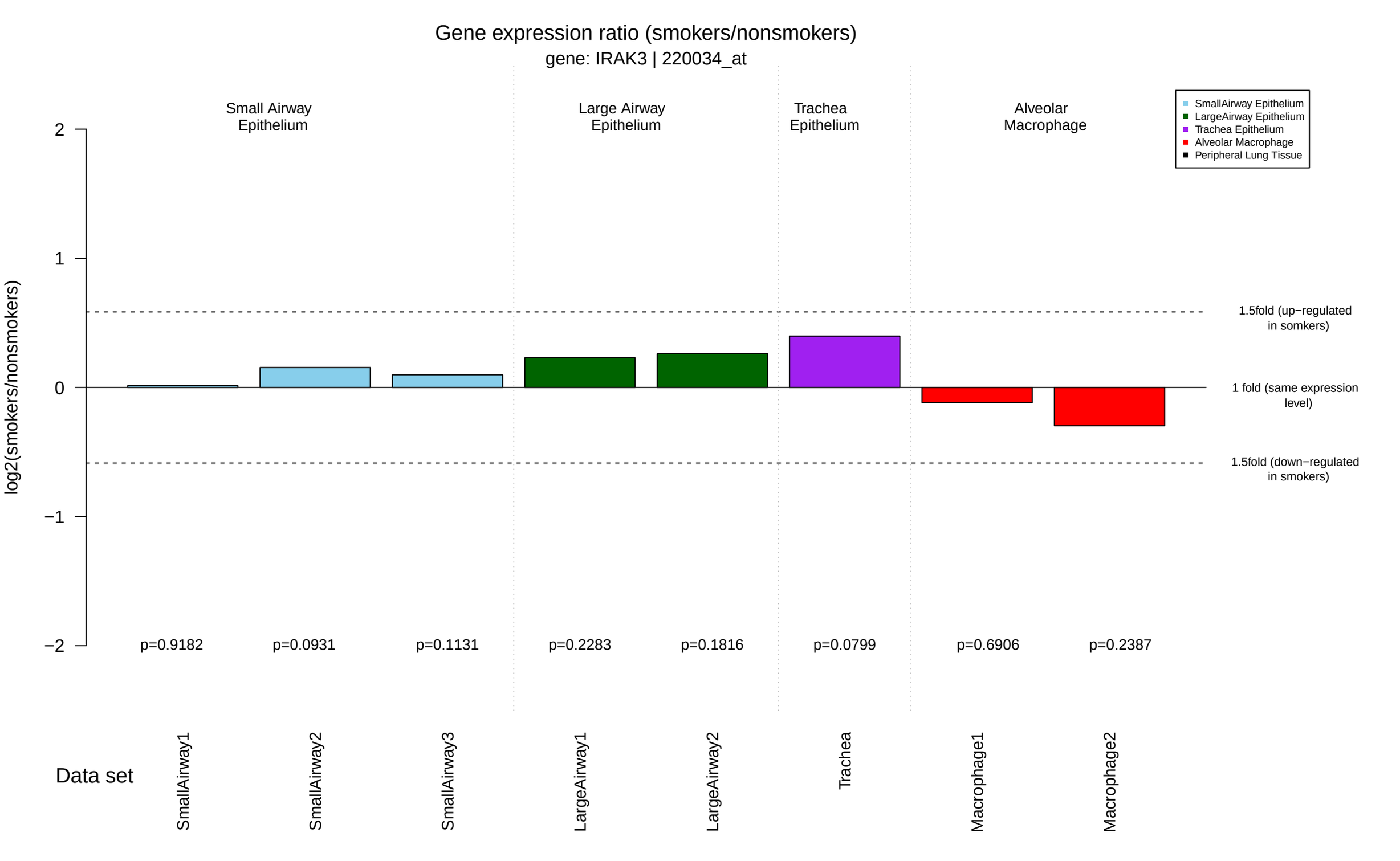
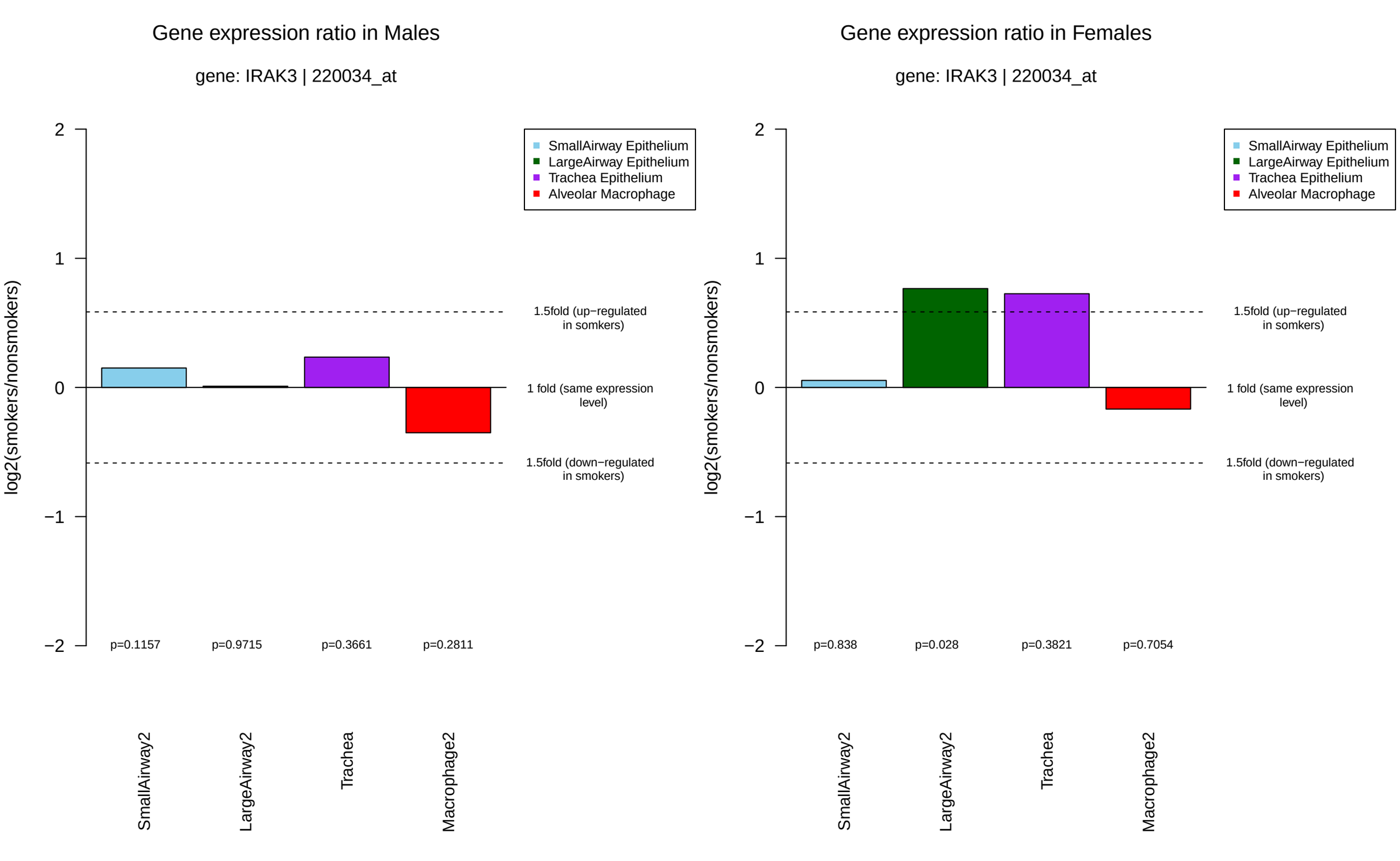
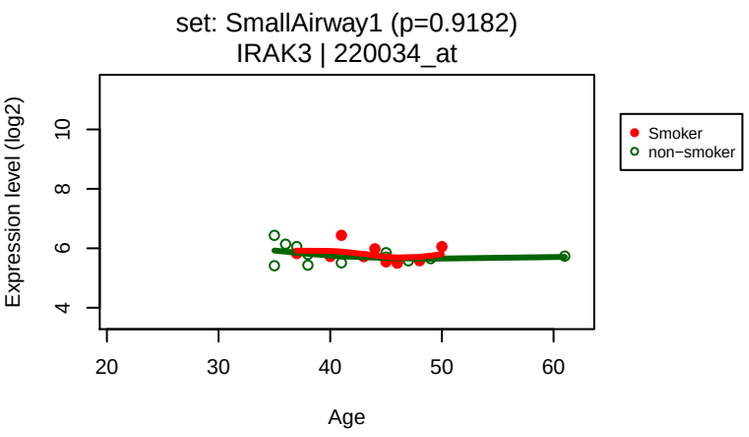
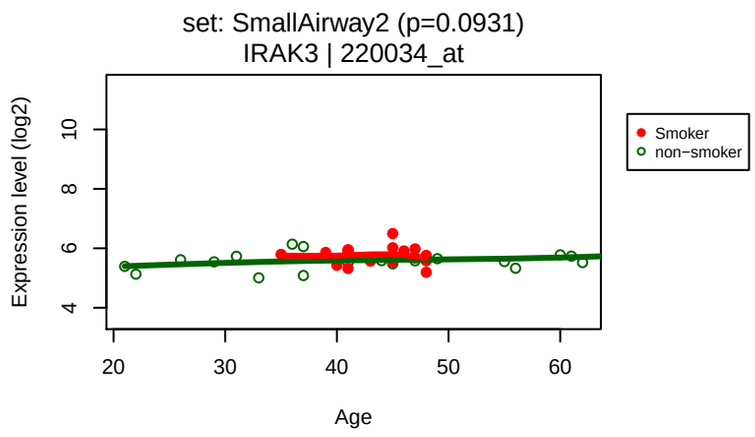
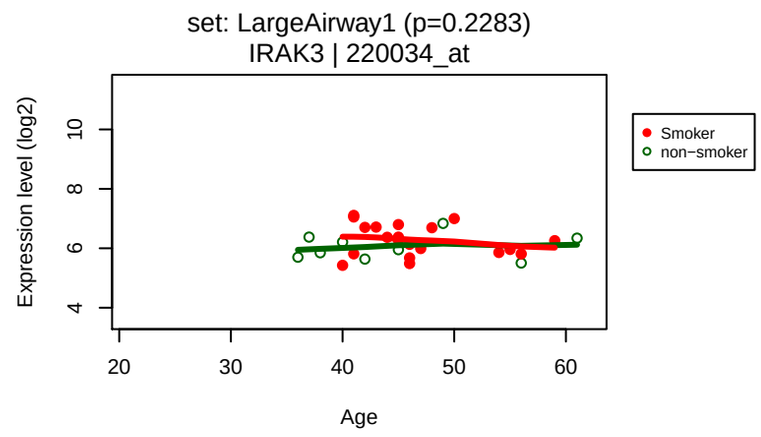

Gene expression of smokers and nonsmokers
gene: IRAK3 | 220034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9182
p=0.0931
p=0.1131
p=0.2283
p=0.1816
p=0.0799
p=0.6906
p=0.2387
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

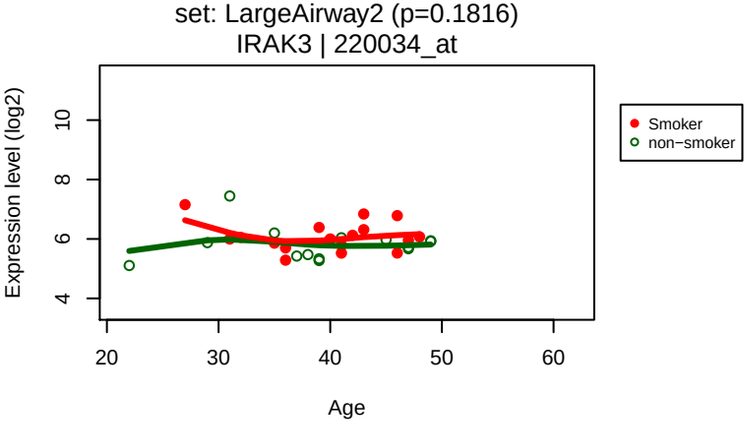

Gene expression ratio (smokers/nonsmokers)
gene: IRAK3 | 220034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9182
p=0.0931
p=0.1131
p=0.2283
p=0.1816
p=0.0799
p=0.6906
p=0.2387
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

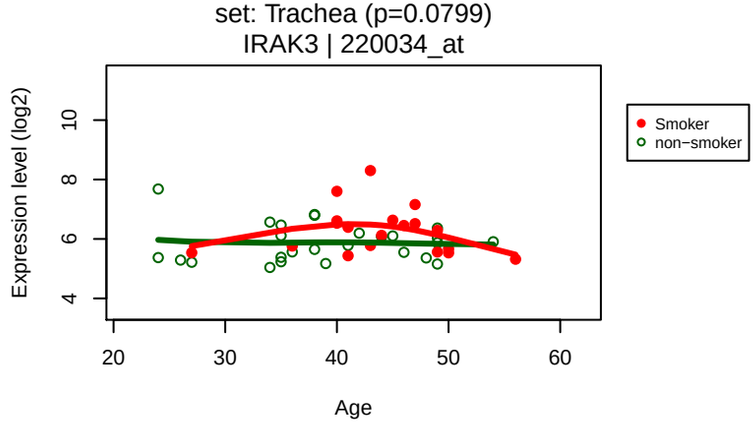

Gene expression ratio in Males
gene: IRAK3 | 220034_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1157
p=0.9715
p=0.3661
p=0.2811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK3 | 220034_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.838
p=0.028
p=0.3821
p=0.7054
SmallAirway2
LargeAirway2
Trachea
Macrophage2

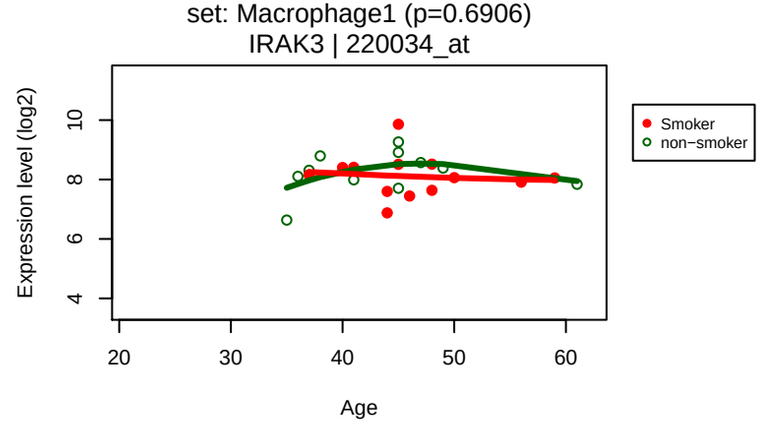

set: SmallAirway1 (p=0.9182)
IRAK3 | 220034_at
Expression level (log2)
Age
Smoker
non−smoker

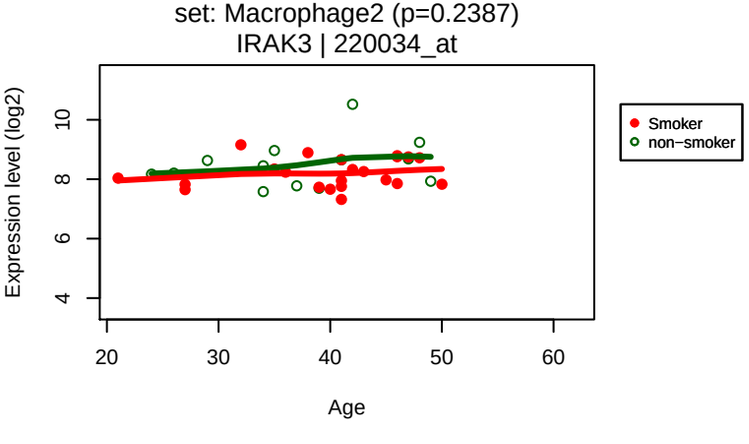

set: SmallAirway2 (p=0.0931)
IRAK3 | 220034_at
Expression level (log2)
Age
Smoker
non−smoker

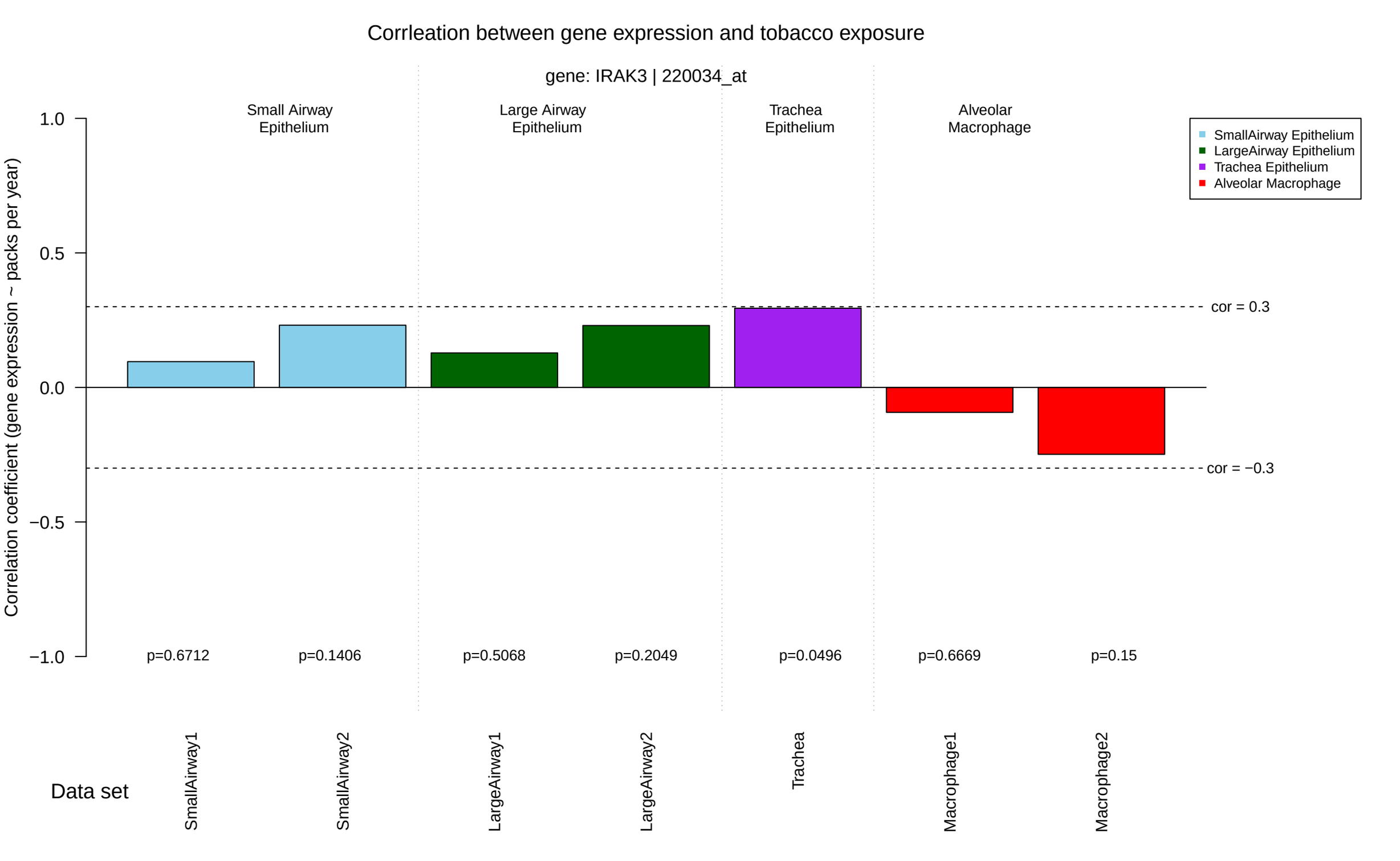

set: LargeAirway1 (p=0.2283)
IRAK3 | 220034_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1816)
IRAK3 | 220034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0799)
IRAK3 | 220034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6906)
IRAK3 | 220034_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2387)
IRAK3 | 220034_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK3 | 220034_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6712
p=0.1406
p=0.5068
p=0.2049
p=0.0496
p=0.6669
p=0.15
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2